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- **Von Mayer** (1845) – Green plants convert solar energy into chemical energy of organic matter.

$$CO_2 + H_2O \longrightarrow \text{Organic matter} + O_2$$

- **Liebig** (1845) – Organic matter of plants was derived from CO_2.
- **Julius Von Sachs** (1854) – Discovered that product of photosynthesis was starch. Green substance (chlorophyll) is located in special structures (Chloroplast).
- **T.W. Engelmann** (1888)- Plotted action spectrum of photosynthesis
- **Blackman** (1905) – Proposed Law of Limiting factors.
- **Warburg** (1920) – Used unicellular green algae *Chlorella* for the study of Photosynthesis.
- **Van Neil** (1931) – Oxygen released during photolysis comes from water and not from CO_2. He also conducted experiments in Purple green bacteria and demonstrated Photosynthesis.

$$2H_2A + CO_2 \xrightarrow[\text{Chlorophyll}]{\text{Light}} (CH_2O)_n + H_2O + 2A$$

In Green Sulphur bacteria H_2S is the Hydrogen donor which releases Sulphur instead of oxygen.

- **Emerson and Arnold** (1932) – Existence of light and dark reaction by flashing light experiments.
- **R. Hill** (1937) – Explained photolysis with the help of isolated chloroplasts and electron acceptors in the presence of light.
- **Ruben and Kamen** (1941) – Used ^{18}O radioactive Oxygen to prove that oxygen evolves from water.
- **Arnon, Allen and Whatley** (1954) – Used radioactive $^{14}CO_2$ to show fixation of CO_2 by isolated chloroplast.
- **Melvin Calvin** (1954) – Used radioactive $^{14}CO_2$ and traced path of carbon in the dark phase of photosynthesis or C_3 Cycle.
- **Emerson** *et al.*, (1957) – Reported existence of two photosystems
- **Hatch and Slack** (1965) – Reported C_4 pathway and CO_2 fixation in C_4 plants
- **Huber, Michel and Dissenhofer** (1985) – Crystalized photosynthetic reaction centre of *Rhodobacter* and received the Nobel Prize in 1988.

13.2 Definition, Significance and Site of Photosynthesis

13.2.1 Definition of Photosynthesis

Photosynthesis is referred as photochemical oxidation and reduction reactions carried out with help of light, converting solar energy into Chemical energy. It is the most important anabolic process. Plants and photosynthetic bacteria use simple raw materials like carbon dioxide water and with the help of light energy synthesize carbohydrates and evolve oxygen. The overall chemical equation for photosynthesis is:

$$6CO_2 + 6H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light}} C_6H_{12}O_6 + 6O_2\uparrow$$

Ruben and **Kamen** (1941) demonstrated six molecules of water as insufficient for the evolution of 6 molecules of O_2 and modified the equation as:

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light}} C_6H_{12}O_6 + 6H_2O + 6O_2\uparrow$$

Photosynthesis is a collection of oxidation and reduction reactions (Redox reaction).

Oxidation- Water is oxidised into oxygen (loss of electrons).